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Université: Institut Universitaire des Sciences (IUS)

TD N° 6: Administration Systèmes Informatiques

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Travaux Dirigés

E-commerce HA + Paiement simulé

1. Contexte

La startup DevCloud souhaite mettre en place une boutique en ligne accessible en permanence (haute disponibilité) avec un système de paiement simulé.

Introduction :

Dans le cadre de ce travail pratique, il s'agit de concevoir et déployer une application e-commerce simple basée sur une architecture à haute disponibilité. Le projet consiste à mettre en place un serveur web utilisant **Nginx** et **PHP-FPM**, puis à développer les fonctionnalités essentielles d'une boutique en ligne telles que le catalogue de produits, le panier d'achat, la validation des commandes et un système de paiement simulé.

2. Objectifs

Développer un site e-commerce complet

Implémenter :

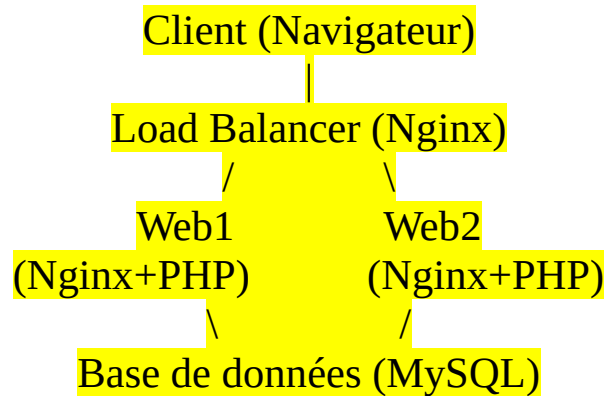
- **Catalogue de produits**
- **Panier**
- **Validation de commande**

Ajouter un bouton "Payer" (simulation)

Mettre en place un **Load Balancer**

Assurer la **haute disponibilité (HA)**

3. Architecture du système



4. Installation des services

a) Installer Nginx et PHP-FPM

sudo apt update

sudo apt install nginx php-fpm -y

```

configure arguments: --with-cc-opt='-g -O2 -fno-omit-frame-pointer -mno-omit-leaf-frame-pointer -ffile-prefix-map=/build/nginx-gpk0Wo/nginx-1.24.0=. -fstack-protector-strong -fstack-clash-protection -Wformat -Werror=format-security -fcf-protection -fdebug-prefix-map=/build/nginx-gpk0Wo/nginx-1.24.0=/usr/src/nginx-1.24.0-2ubuntu7.6 -fPIC -Wdate-time -D_FORTIFY_SOURCE=3' --with-ld-opt='-Wl,-Bsymbolic-functions -flto=auto -ffat-lto-objects -Wl,-z,relro -Wl,-z,now -fPIC' --prefix=/usr/share/nginx --conf-path=/etc/nginx/nginx.conf --http-log-path=/var/log/nginx/access.log --error-log-path=/dev/stderr --lock-path=/var/lock/nginx.lock --pid-path=/run/nginx.pid --modules-path=/usr/lib/nginx/modules --http-client-body-temp-path=/var/lib/nginx/body --http-fastcgi-temp-path=/var/lib/nginx/fastcgi --http-proxy-temp-path=/var/lib/nginx/proxy --http-scgi-temp-path=/var/lib/nginx/scgi --http-uwsgi-temp-path=/var/lib/nginx/uwsgi --with-compat --with-debug --with-pcre-jit --with-http_ssl_module --with-http_stub_status_module --with-http_realip_module --with-http_auth_request_module --with-http_v2_module --with-http_dav_module --with-http_slice_module --with-threads --with-http_addition_module --with-http_flv_module --with-http_gunzip_module --with-http_gzip_static_module --with-http_mp4_module --with-http_random_index_module --with-http_secure_link_module --with-http_sub_module --with-mail_ssl_module --with-stream_ssl_module --with-stream_ssl_preread_module --with-stream_realip_module --with-http_geoip_module=dynamic --with-http_image_filter_module=dynamic --with-http_perl_module=dynamic --with-http_xslt_module=dynamic --with-mail=dynamic --with-stream=dynamic --with-stream_geoip_module=dynamic
evenort12@ubuntu:~$ sudo apt update
nginx version: nginx/1.24.0 (Ubuntu)
evenort12@ubuntu:~$ php -v
PHP 8.3.6 (cli) (built: Mar 20 2026 02:32:55) (NTS)
Copyright (c) The PHP Group
Zend Engine v4.3.6, Copyright (c) Zend Technologies
    with Zend OPcache v8.3.6, Copyright (c), by Zend Technologies
evenort12@ubuntu:~$ php-fpm -v
php-fpm: command not found
evenort12@ubuntu:~$ php-fpm -v
php-fpm: command not found
evenort12@ubuntu:~$ apt list --installed | grep nginx
WARNING: apt does not have a stable CLI interface. Use with caution in scripts.

nginx-common/noble-updates,noble-security,now 1.24.0-2ubuntu7.6 all [installé, automatique]
nginx/noble-updates,noble-security,now 1.24.0-2ubuntu7.6 amd64 [installé]
evenort12@ubuntu:~$ apt list --installed | grep php
WARNING: apt does not have a stable CLI interface. Use with caution in scripts.

php-common/noble,now 2:93ubuntu2 all [installé, automatique]
php-fpm/noble,now 2:8.3+93ubuntu2 all [installé]
php8.3-cli/noble-updates,now 8.3.6-0ubuntu0.24.04.8 amd64 [installé, automatique]
php8.3-common/noble-updates,now 8.3.6-0ubuntu0.24.04.8 amd64 [installé, automatique]
php8.3-fpm/noble-updates,now 8.3.6-0ubuntu0.24.04.8 amd64 [installé, automatique]
php8.3-opcache/noble-updates,now 8.3.6-0ubuntu0.24.04.8 amd64 [installé, automatique]
php8.3-readline/noble-updates,now 8.3.6-0ubuntu0.24.04.8 amd64 [installé, automatique]
evenort12@ubuntu:~$
  
```

Figure 1: Cette image illustre l'installation de Nginx et PHP ainsi que la vérification de leur état.

b) Sur chaque serveur web :

sudo apt update

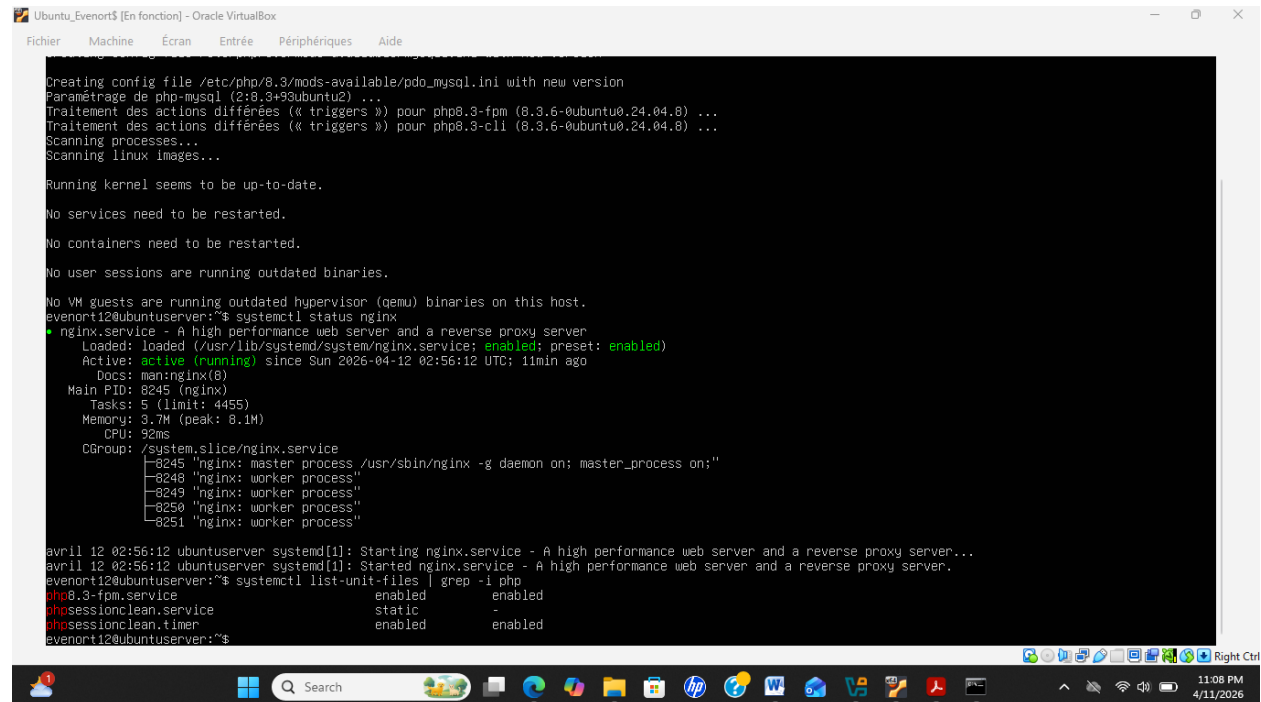
sudo apt install nginx php-fpm php-mysql -y

c) Vérification:

systemctl status nginx

systemctl list-unit-files | grep -i php

systemctl status php8.4-fpm



```
Creating config file /etc/php/8.3/mods-available/pdo_mysql.ini with new version
Paramétrage de php-mysql (2:8.3+93ubuntu2) ...
Traitement des actions différées (« triggers ») pour php8.3-fpm (8.3.6-0ubuntu0.24.04.8) ...
Traitement des actions différées (« triggers ») pour php8.3-cli (8.3.6-0ubuntu0.24.04.8) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
evenort12@ubuntu:~$ systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-04-12 02:56:12 UTC; 11min ago
     Docs: man:nginx(8)
   Main PID: 8245 (nginx)
      Tasks: 5 (limit: 4455)
    Memory: 3.7M (peak: 8.1M)
       CPU: 92ms
   CGroup: /system.slice/nginx.service
           └─8245 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─8248 "nginx: worker process"
               └─8249 "nginx: worker process"
                 └─8250 "nginx: worker process"
                   └─8251 "nginx: worker process"

avril 12 02:56:12 ubuntu:server systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
avril 12 02:56:12 ubuntu:server systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
evenort12@ubuntu:~$ systemctl list-unit-files | grep -i php
php8.3-fpm.service          enabled          enabled
phpsessionclean.service    static          -
phpsessionclean.timer      enabled         enabled
evenort12@ubuntu:~$
```

Figure 2: Cette capture d'écran démontre l'état opérationnel de **Nginx** et de **PHP**.

5. Développement du site

a) Création du dossier web

sudo mkdir -p /var/www/html

b) Accéder au dossier web

cd /var/www/html

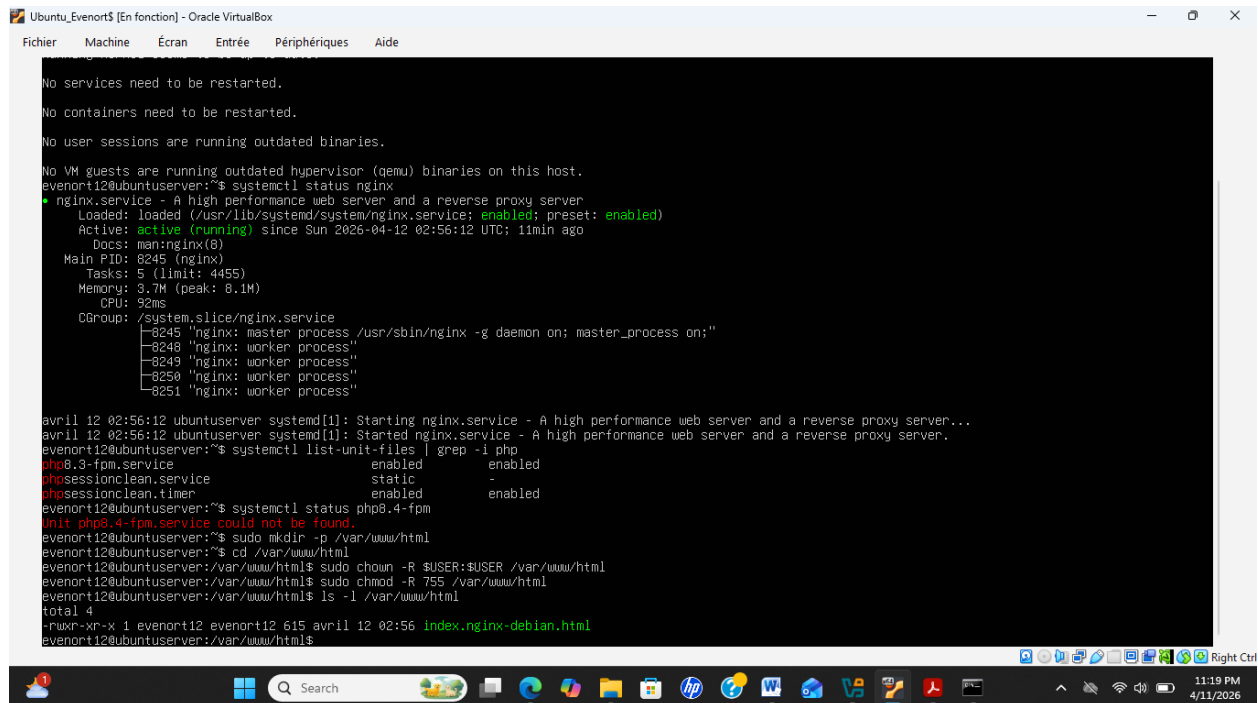
c) Donner les permissions

sudo chown -R \$USER:\$USER /var/www/html

sudo chmod -R 755 /var/www/html

c) Vérification

ls -l /var/www/html



```

No services need to be restarted.
No containers need to be restarted.
No user sessions are running outdated binaries.
No VM guests are running outdated hypervisor (qemu) binaries on this host.
evenort12@ubuntu:~$ systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-04-12 02:56:12 UTC; 11min ago
     Docs: man:nginx(8)
    Main PID: 8245 (nginx)
      Tasks: 5 (limit: 4455)
    Memory: 3.7M (peak: 8.1M)
       CPU: 92ms
    CGroup: /system.slice/nginx.service
            └─8245 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
              └─8248 "nginx: worker process"
                └─8249 "nginx: worker process"
                  └─8250 "nginx: worker process"
                    └─8251 "nginx: worker process"

avril 12 02:56:12 ubuntu:systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
avril 12 02:56:12 ubuntu:systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
evenort12@ubuntu:~$ systemctl list-unit-files | grep -i php
php8.3-fpm.service          enabled          enabled
phpsessionclean.service    static          -
phpsessionclean.timer      enabled          enabled
evenort12@ubuntu:~$ systemctl status php8.4-fpm
Unit php8.4-fpm.service could not be found.
evenort12@ubuntu:~$ sudo mkdir -p /var/www/html
evenort12@ubuntu:~$ cd /var/www/html
evenort12@ubuntu:~/var/www/html$ sudo chown -R $USER:$USER /var/www/html
evenort12@ubuntu:~/var/www/html$ sudo chmod -R 755 /var/www/html
evenort12@ubuntu:~/var/www/html$ ls -l /var/www/html
total 4
-rwxr-xr-x 1 evenort12 evenort12 615 avril 12 02:56 index.nginx-debian.html
evenort12@ubuntu:~/var/www/html$
```

Figure 3: Cette capture d'écran montre la création du répertoire web et la vérification des permissions accordées via la commande `ls -l /var/www/html`.

6. Création des fichiers PHP

a) index.php

`sudo nano /var/www/html/index.php`

```
<?php
```

```
session_start();
```

```
$produits = [
```

```
    ["id"=>1, "nom"=>"Laptop", "prix"=>800],
```

```
    ["id"=>2, "nom"=>"Téléphone", "prix"=>300],
```

```
];
```

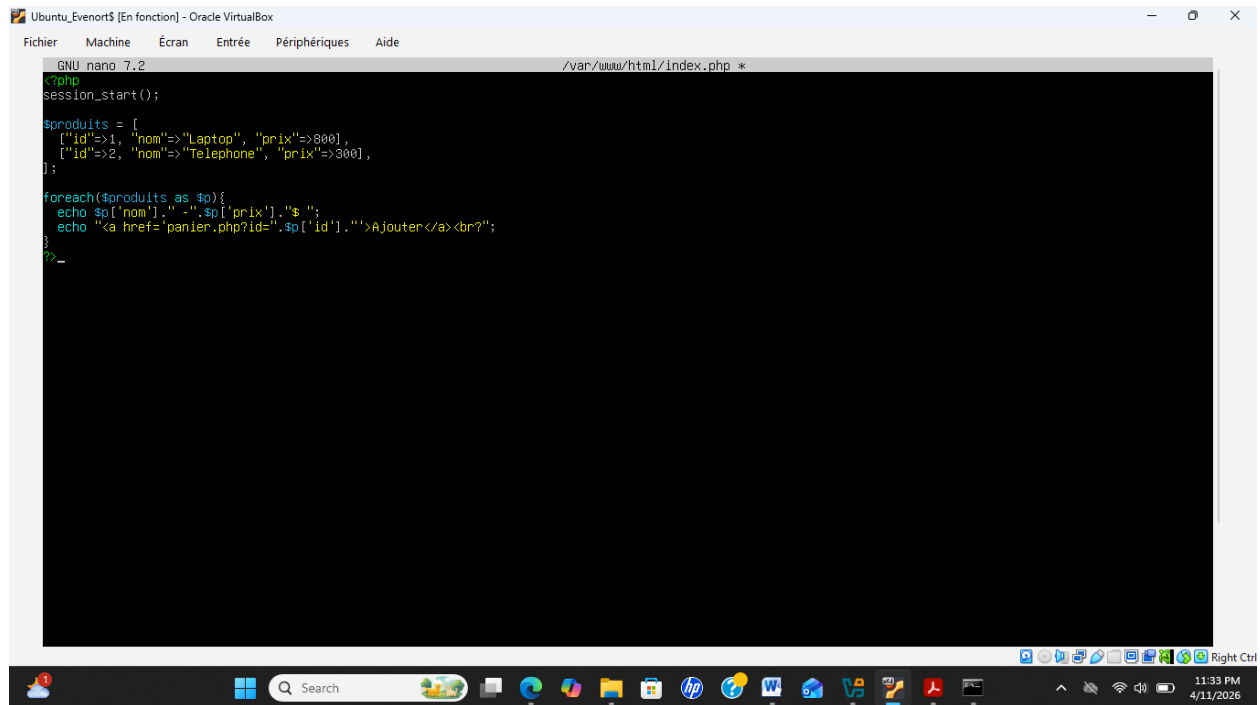
```
foreach($produits as $p){
```

```
    echo $p['nom']. " - ".$p['prix']."$ ";
```

```
    echo "<a href='panier.php?id=".$p['id']."'>Ajouter</a><br>";
```

```
}
```

```
?>
```

A screenshot of a terminal window titled 'Ubuntu_Evenor\$ [En fonction] - Oracle VirtualBox'. The terminal shows the command 'nano /var/www/html/index.php' being executed. The nano editor interface displays the following PHP code:

```
<?php
session_start();

$produits = [
    ['id'=>1, 'nom'=>'Laptop', 'prix'=>800],
    ['id'=>2, 'nom'=>'Telephone', 'prix'=>300],
];

foreach($produits as $p){
    echo $p['nom']. " - " . $p['prix']. "$ ";
    echo "<a href='panier.php?id=".$p['id']."'>Ajouter</a><br?";
}
?>
```

The terminal window has a menu bar with 'Fichier', 'Machine', 'Écran', 'Entrée', 'Périphériques', and 'Aide'. The status bar at the bottom shows the time as 11:33 PM on 4/11/2026.

Figure 4: Cette capture d'écran atteste la création du fichier **index.php**.

b) panier.php

sudo nano /var/www/html/panier.php

```
<?php
```

```
session_start();
```

```
$id = $_GET['id'];
```

```
$_SESSION['panier'][] = $id;
```

```
echo "Produit ajouté au panier<br>";
```

```
echo "<a href='commande.php'>Voir commande</a>";
```

```
?>
```

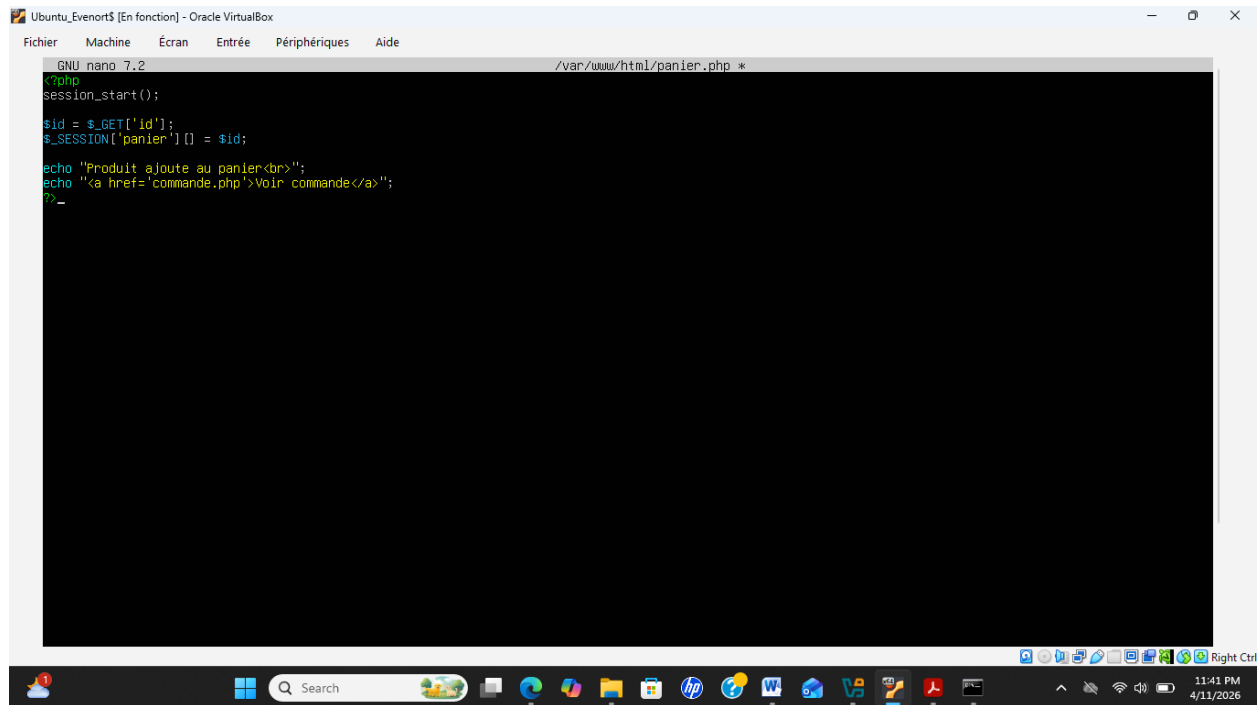


Figure 5: Cette capture d'écran illustre la création du fichier **panier.php**.

c) commande.php

```
sudo nano /var/www/html/commande.php
```

```
<?php
```

```
session_start();
```

```
echo "Produits dans le panier:<br>";
```

```
foreach($_SESSION['panier'] as $id){
```

```
    echo "Produit ID: ".$id."<br>";
```

```
}
```

```
echo "<a href='payer.php'>Payer</a>";
```

```
?>
```

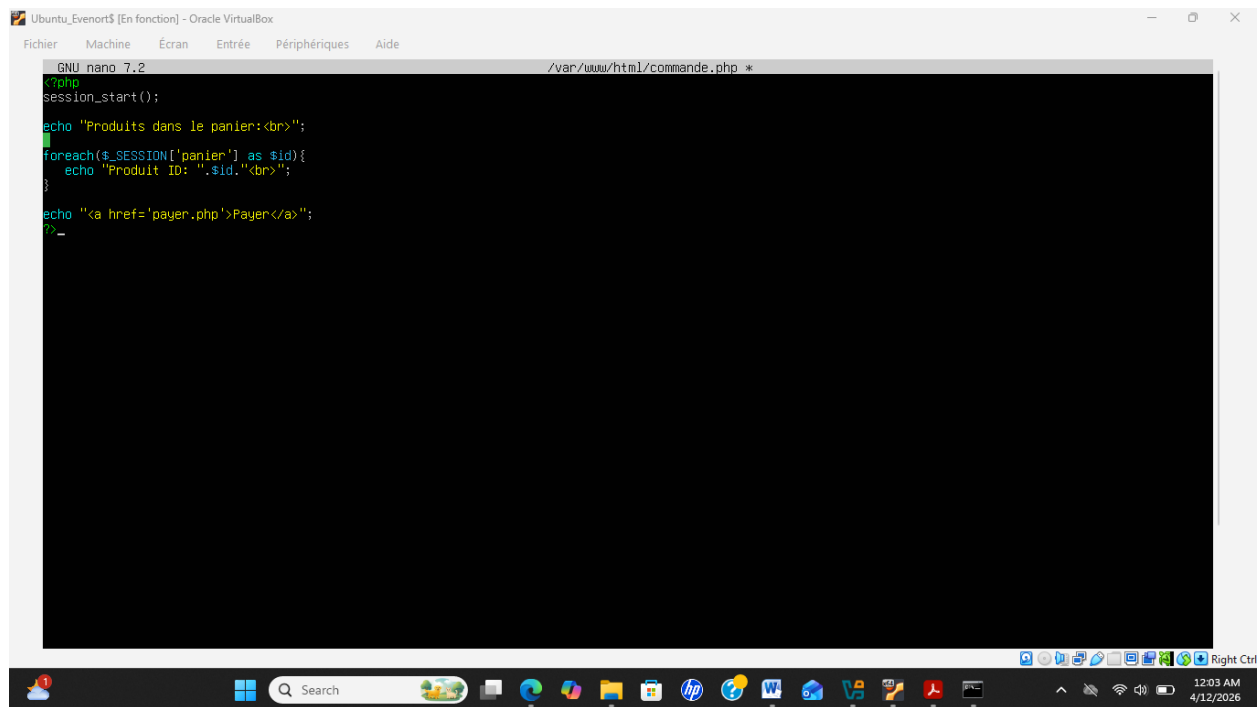


Figure 6: Cette capture d'écran illustre la création du fichier **commande.php**.

d) payer.php

sudo nano /var/www/html/payer.php

```
<?php
```

```
session_start();
```

```
echo "Produits dans le panier:<br>";
```

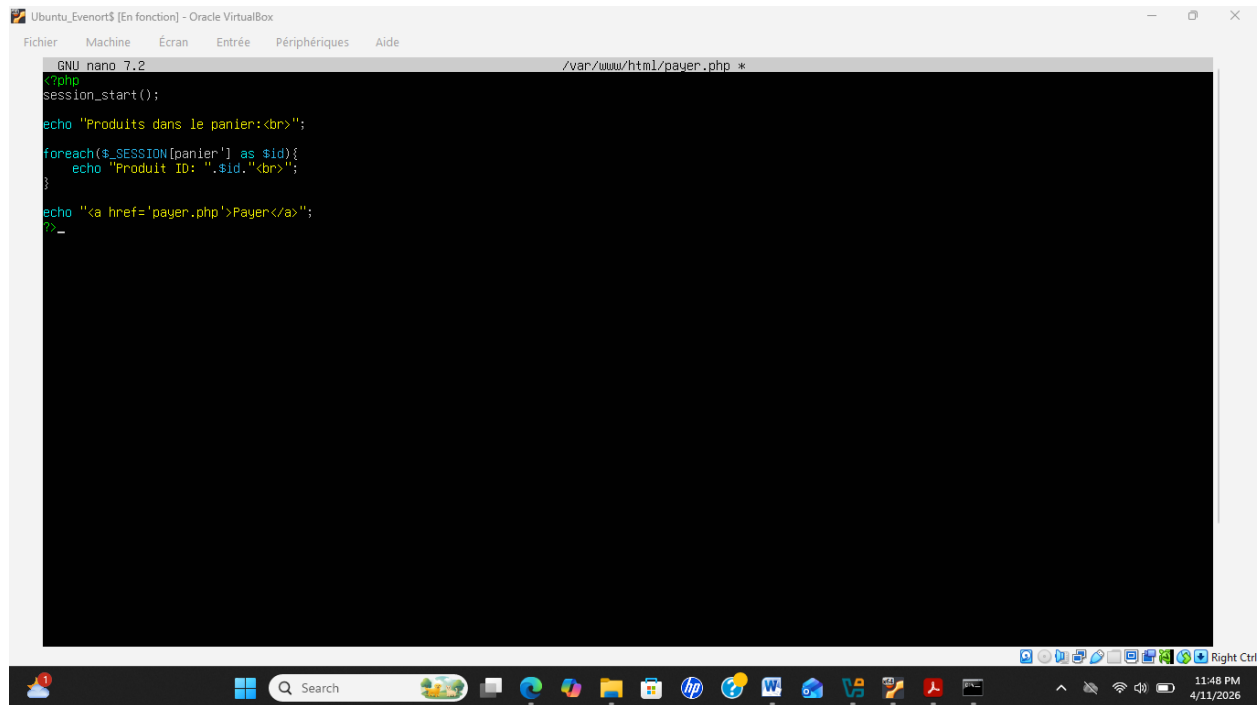
```
foreach($_SESSION['panier'] as $id){
```

```
    echo "Produit ID: ".$id."<br>";
```

```
}
```

```
echo "<a href='payer.php'>Payer</a>";
```

```
?>
```

```
GNU nano 7.2 /var/www/html/payer.php *
<?php
session_start();

echo "Produits dans le panier:<br>";
foreach($_SESSION['panier'] as $id){
    echo "Produit ID: ".$id."<br>";
}

echo "<a href='payer.php'>Payer</a>";
?>
```

Figure 7: Cette capture d'écran illustre la création du fichier **payer.php**.

Permissions:

```
sudo chown -R $USER:$USER /var/www/html
```

```
sudo chmod -R 755 /var/www/html
```

6. Configuration PHP avec Nginx

```
sudo nano /etc/nginx/sites-available/default
```

a) ajouter:

```
server {
```

```
    listen 80;
```

```
    listen [::]:80;
```

```
    server_name _;
```

```
    root /var/www/html;
```

```
    index index.php index.html;
```

```
    location / {
```

```
        try_files $uri $uri/ =404;
```

```
    }
```

```
location ~ \.php$ {  
    include snippets/fastcgi-php.conf;  
    fastcgi_pass unix:/var/run/php/php8.1-fpm.sock;  
}
```

```
location ~ /\.ht {  
    deny all;  
}
```

```
}
```

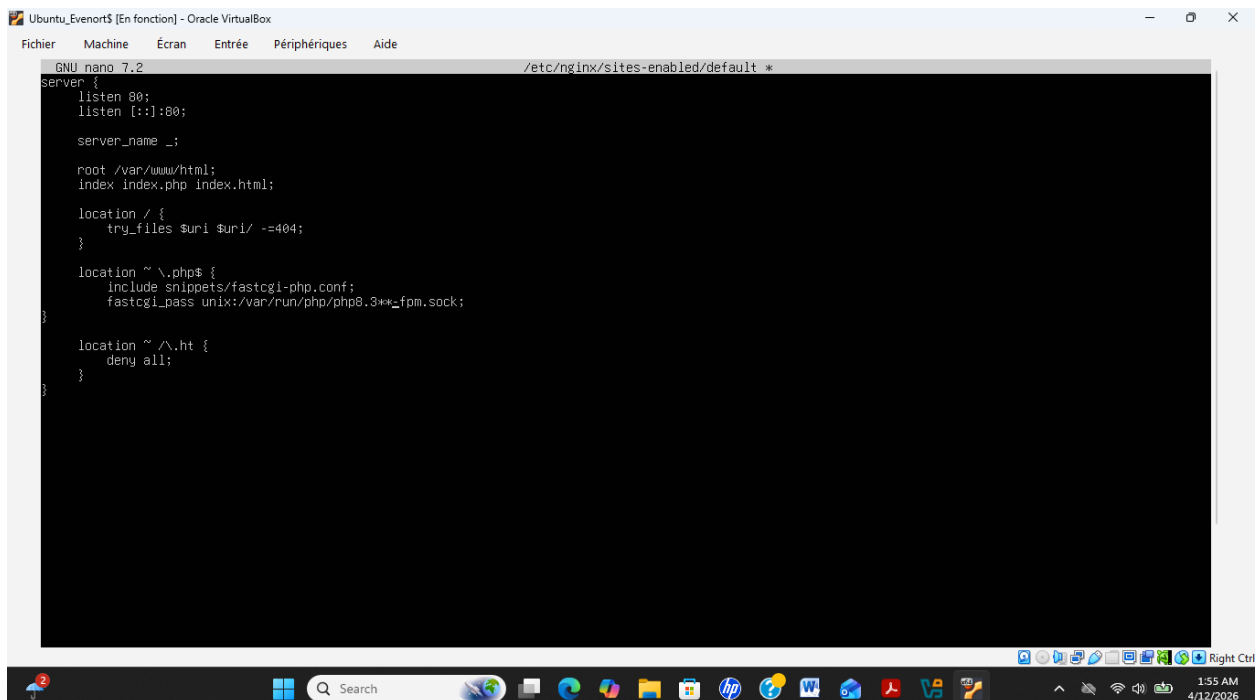


Figure 8: Cette capture d'écran illustre la configuration du bloc **serveur Nginx** pour le support de **PHP**.

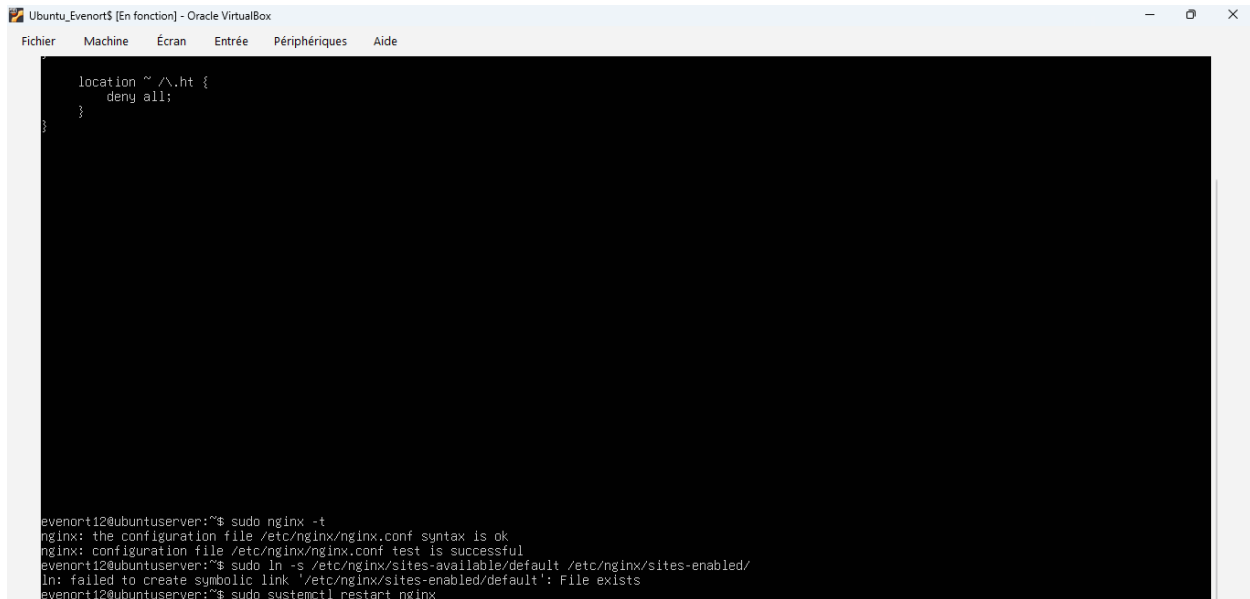
b) **Vérification** : `sudo nginx -t`

7. Activer le site

`sudo ln -s /etc/nginx/sites-available/default /etc/nginx/sites-enabled/`

a) **Redémarrer**

`sudo systemctl restart nginx`



```
location ~ /\.ht {
    deny all;
}

evenort12@ubuntu:~$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
evenort12@ubuntu:~$ sudo ln -s /etc/nginx/sites-enabled/default /etc/nginx/sites-enabled/
ln: failed to create symbolic link '/etc/nginx/sites-enabled/default': File exists
evenort12@ubuntu:~$ sudo systemctl restart nginx
```

Figure 9: Cette capture d'écran montre la vérification de la configuration (**sudo nginx -t**) suivie de l'activation du site.

8. Load Balancer avec Nginx

sudo nano /etc/nginx/nginx.conf

a) Ajouter :

upstream backend {

server 192.168.56.103;

server 192.168.56.104;

}

server {

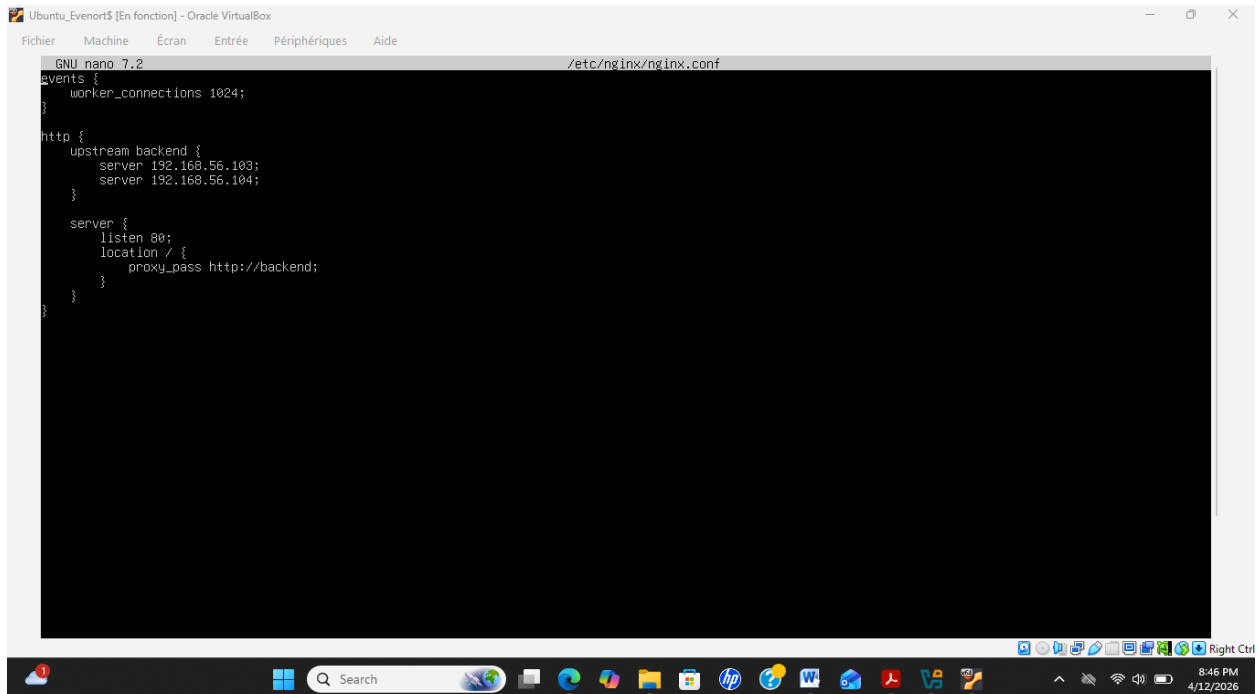
listen 80;

location / {

proxy_pass http://backend;

}

}



```
GNU nano 7.2 /etc/nginx/nginx.conf
events {
    worker_connections 1024;
}

http {
    upstream backend {
        server 192.168.56.103;
        server 192.168.56.104;
    }

    server {
        listen 80;
        location / {
            proxy_pass http://backend;
        }
    }
}
```

Figure 10: Cette capture d'écran illustre la configuration du répartiteur de charge (**Load Balancer**) avec **Nginx**.

b) Redémarrage

sudo systemctl restart nginx

9. Test Haute Disponibilité

a) Accéder à: http://192.168.56.103_LoadBalancer

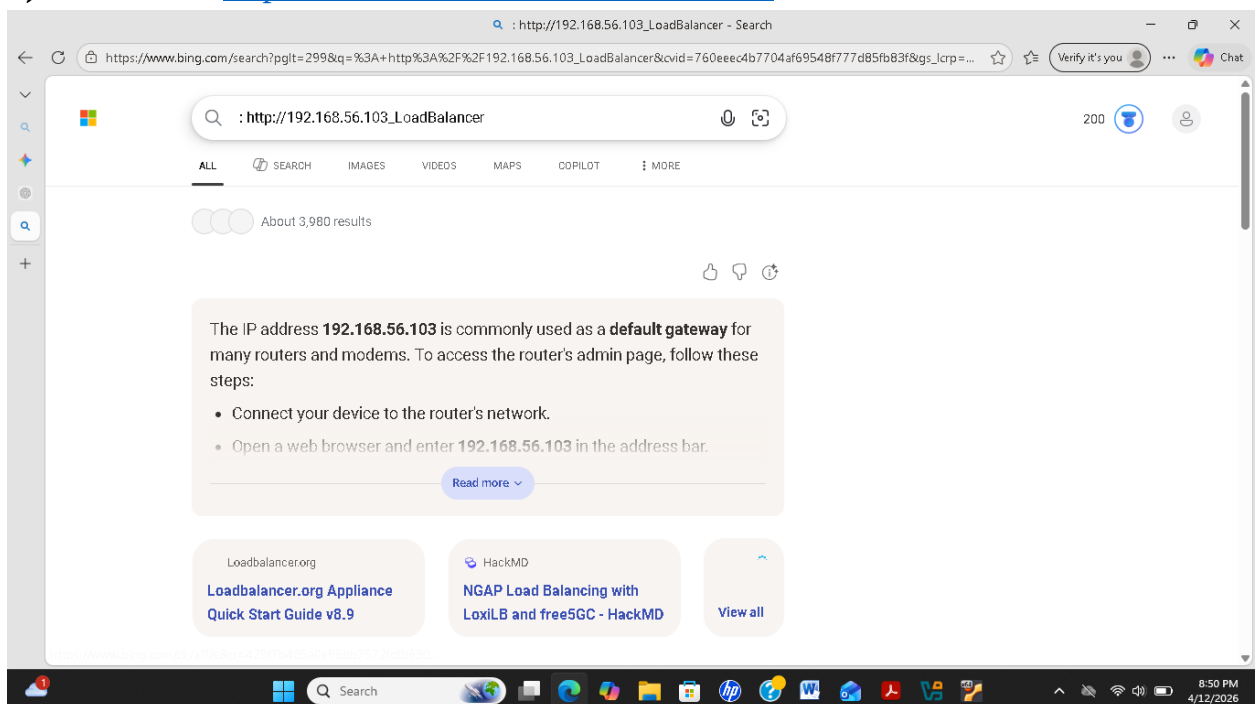


Figure 11: Cette capture d'écran illustre le test de Haute Disponibilité, lequel a été réussi avec succès.

b) Test panne serveur

`sudo systemctl stop nginx`

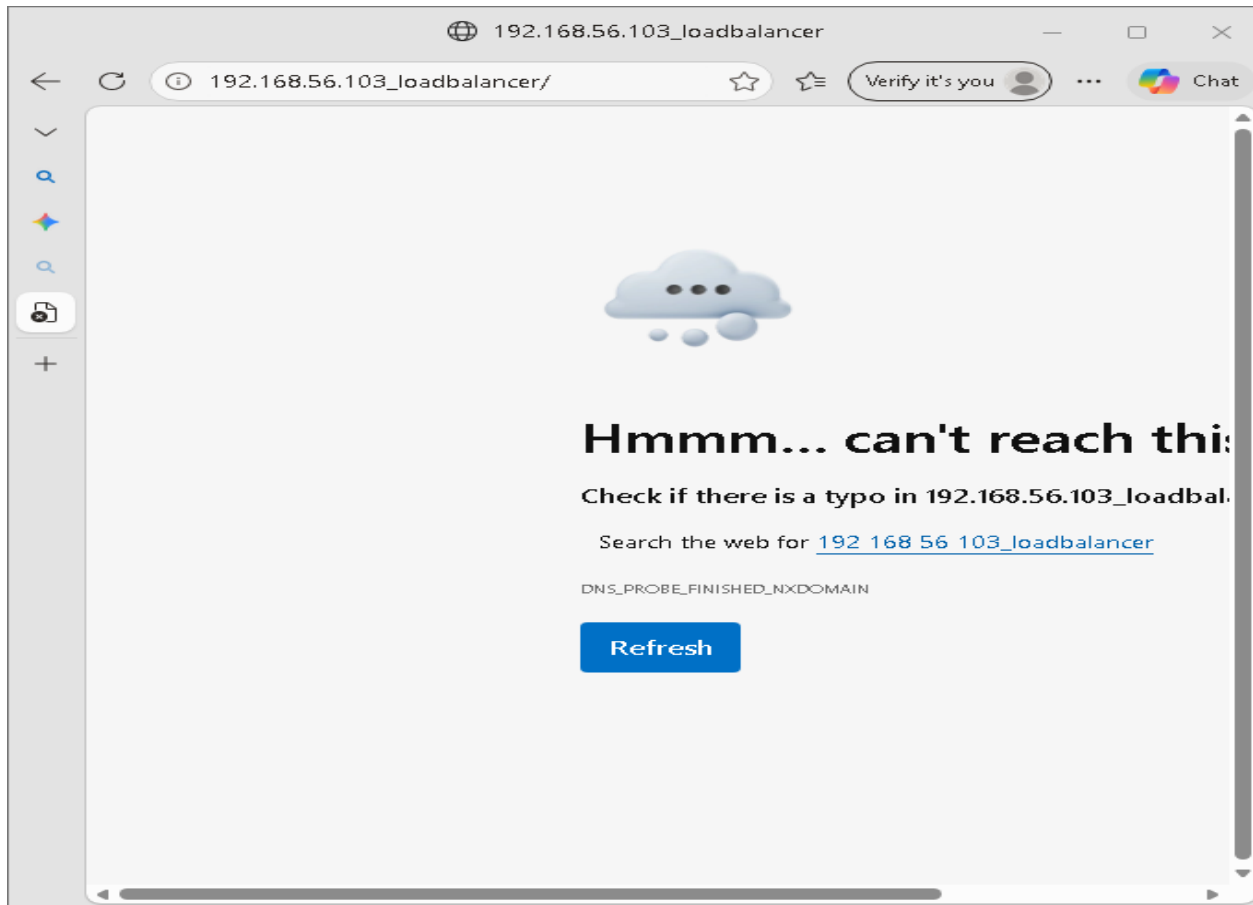


Figure 12: Dans cette image, on constate l'arrêt du **service Nginx** sur le **load balancer** principal. Par conséquent, le site devient indisponible et présente un point unique de défaillance.

Conclusion

En conclusion, ce travail a permis de mettre en œuvre les bases du développement d'une application e-commerce tout en intégrant des notions d'administration système et réseau. L'utilisation de Nginx comme serveur web et répartiteur de charge a permis d'assurer la disponibilité du service même en cas de panne.